

Grand Challenges for Distributed Mixed Reality Collaboration: A Research Agenda

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Jens Emil Grønbæk

Aarhus University, Denmark
jensemil@cs.au.dk

Adélaïde Genay

University of Melbourne, Australia
adelaide.genay@unimelb.edu.au

Andrew Irlitti

Deakin University, Australia
andrew.irlitti@deakin.edu.au

Brandon Victor Syiem

University of Sydney, Australia
brandon.syiem@sydney.edu.au

Jarrod Knibbe

University of Queensland, Australia
j.knibbe@uq.edu.au

Emily Wong

University of Copenhagen, Denmark
emily.wong@di.ku.dk

Tiare Feuchtner

University of Konstanz, Germany
tiare.feuchtner@uni-konstanz.de

Tinghui Li

University of Sydney, Australia
tinghui.li@sydney.edu.au

Zhongyi Bai

University of Sydney, Australia
zhongyi.bai@sydney.edu.au

Eduardo Velloso

University of Sydney, Australia
eduardo.velloso@sydney.edu.au

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Jens Emil Grønbaek¹, Adélaïde Genay², Andrew Irlitti³, Brandon Victor Syiem⁴, Jarrod Knibbe⁵, Emily Wong⁶, Tiare Feuchtner⁷, Tinghui Li⁸, Zhongyi Bai⁹ and Eduardo Velloso¹⁰

¹*Aarhus University, Denmark; jensemil@cs.au.dk*

²*University of Melbourne, Australia; adelaide.genay@unimelb.edu.au*

³*Deakin University, Australia; andrew.irlitti@deakin.edu.au*

⁴*University of Sydney, Australia; brandon.syiem@sydney.edu.au*

⁵*University of Queensland, Australia; j.knibbe@uq.edu.au*

⁶*University of Copenhagen, Denmark; emily.wong@di.ku.dk*

⁷*University of Konstanz, Germany; tiare.feuchtner@uni-konstanz.de*

⁸*University of Sydney, Australia; tinghui.li@sydney.edu.au*

⁹*University of Sydney, Australia; zhongyi.bai@sydney.edu.au*

¹⁰*University of Sydney, Australia; eduardo.velloso@sydney.edu.au*

ABSTRACT

In this paper, we unpack one of the key promises of Mixed Reality (MR) — enabling distributed users to collaborate as if they were face-to-face, while each user’s interactions remain grounded in their physical reality. Decades of research have highlighted the expressive capabilities of MR, but we have still not solved the fundamental challenges to enable distributed team collaboration in a modern work context. To articulate these fundamentals, we outline a research agenda

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for MR interaction in modern distributed team collaboration. We present six grand challenges, based on tensions between principles of face-to-face interaction and the demands of modern work. To advance efforts, we take stock of state-of-the-art MR research related to each challenge and identify open research questions for Human-Computer Interaction. Our aim with this manuscript is to provide a future outlook for how to bring collaborative MR to where it is most needed in modern distributed teamwork.

1

Introduction

Imagine if distributed teams could collaborate through embodied and spatial interaction, leveraging their innate abilities for face-to-face interaction. This vision has long been part of the collective imagination, inspiring numerous works of science fiction, such as the hologram technology exhibited in the Star Wars movies.

Decades of research have demonstrated the potential for Mixed Reality (MR) to bring spatial interaction qualities to collaboration across geographic locations (Ens *et al.*, 2019). Unlike fully immersive technologies such as Virtual Reality (VR, where users are not aware of their physical reality), MR blends the physical with the digital. This enables collaborative work to leverage familiar principles of face-to-face interaction, not only through human-human spatial relations but also human-environment relations, as interactions are grounded in physical space. Spatial referencing and physical object manipulation are core examples of such interactions. This grounding in physical reality positions MR as a compelling medium for distributed collaboration, which can create a strong sense of *being there* together (Ditton and Lombard, 1997; Heater, 1992), and ultimately go *beyond* what face-to-face interaction alone affords (Hollan and Stornetta, 1992; McVeigh-

Schultz and Isbister, 2022; Billinghurst and Kato, 1999a; Lunding *et al.*, 2023).

From a technical perspective, we have come a long way in bringing collaborative MR to everyday users. Since the early MR prototypes in the '90s (e.g., Billinghurst and Kato (1999b), Billinghurst and Kato (1999a), and Raskar *et al.* (1998)), MR Head-Mounted Displays (HMDs) have matured significantly and are now at a point where consumer adoption is feasible (Ens *et al.*, 2019). While there is still a considerable gap between the vision and today's MR hardware, new headsets such as Meta Quest or Apple Vision Pro offer an unprecedented opportunity for MR to provide real value in distributed work, by reducing the time and environmental footprint of travelling around for face-to-face collaboration.

Despite these potential benefits, persistent challenges hinder the adoption and practical application of MR for collaboration. We can broadly divide these challenges into two kinds. The first kind is hardware-related, such as technology availability, device form factors, physical setup requirements, and training and familiarity of using a new kind of physical device. Although highly important, the long-term impact of such limitations is largely speculative and contingent on broader hardware advances. The second kind of challenges are the *human-centred interaction challenges*—those concerning the human-to-human experience of using MR as a collaborative medium, which can be studied in principle, independently of any particular hardware trajectory. Figure 1.1 illustrates the nature of such challenges, drawing on the hologram meeting scene from Star Wars as a familiar point of reference. Given that distributed team meetings are often cognitively demanding (Baym *et al.*, 2021), we posit that before MR can go *beyond* what face-to-face interaction offers, it first needs to provide a sufficiently effective sense of *being there* by leveraging our innate skills to interact with each other face-to-face.

This manuscript is a state-of-the-art literature overview and a timely call to action for researchers in the field of Human-Computer Interaction (HCI) working on MR interaction challenges. Based on prior literature, we articulate six human-centred MR interaction challenges of modern distributed team collaboration. For each challenge, we survey the state-

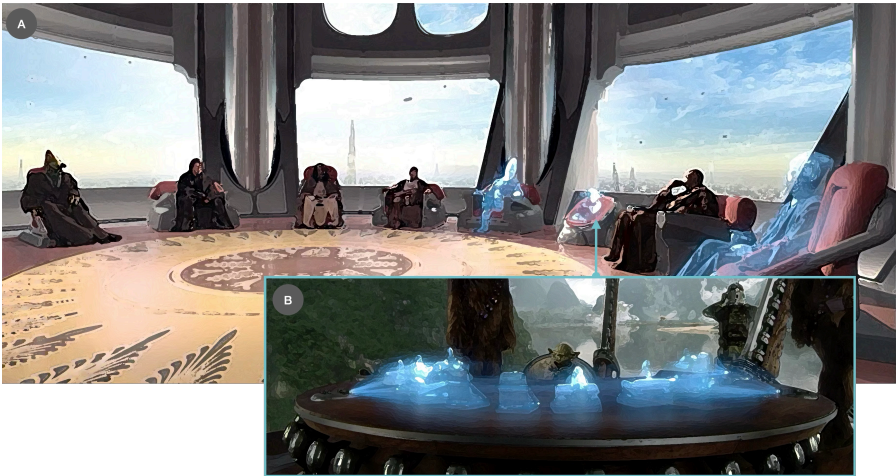


Figure 1.1: Star Wars' famous science fiction depiction of distributed team collaboration using holograms. By juxtaposing two sides of the collaborative experience, it reveals the communication asymmetries that the team must navigate. (A) The team is seated in a circular facing formation, and one team member's hologram conveys the embodied gesture of looking to the side. (B) However, the same team member's local experience and space is completely different, requiring the holograms of the rest of the team to be retrofitted to the local surroundings.

of-the-art research, identify where progress has been made, and discuss potential pathways toward addressing the open questions that remain.

Regardless of the chosen technology, bringing the principles of face-to-face interaction to modern distributed work is far from straightforward. Prior literature on Computer-Supported Cooperative Work (CSCW) has built a well-grounded understanding of effective collaboration in face-to-face settings — particularly important is research that draws upon theories on the spatial dimensions of social interaction (such as proxemics (Hall, 1966) and f-formations (Kendon, 1990)), which serves as the background (see Chapter 2) of our research agenda. However, creating shared realities in MR becomes a completely different challenge when confronted with the communication asymmetries of modern distributed work. Despite a wealth of conceptual work on presence and mediated co-presence (Heater, 1992; Biocca and Harms, 2002; Tran *et al.*, 2024; Wagner *et al.*, 2009; Ens *et al.*, 2019), little research directly confronts the tensions between creating a sense of being together and the messy realities of distributed work.

This manuscript organises these tensions around six grand challenges, each grounded in a fundamental principle of embodied, small-group collaboration identified in prior CSCW literature: *spatially congruent actions*, *embodied interaction*, *organic group interactions*, *live interaction*, *tangibility*, and *fluid task switching*. Each of these principles is a critical characteristic of how groups of collaborators interact in physical environments.

However, these face-to-face principles are at odds with the realities of modern distributed work. Today, people may work from *heterogeneous spaces*, computer-mediated interactions require *user representations*, communication tools struggle to support *group dynamics at scale*, physical work materials are distributed across *task spaces* in different locations and time zones, and collaborating now involves *complex ecologies* of digital devices and applications.

1.1 Contribution: a research agenda

We outline a research agenda for Mixed Reality to bridge the divide between principles of face-to-face interaction and requirements of mod-

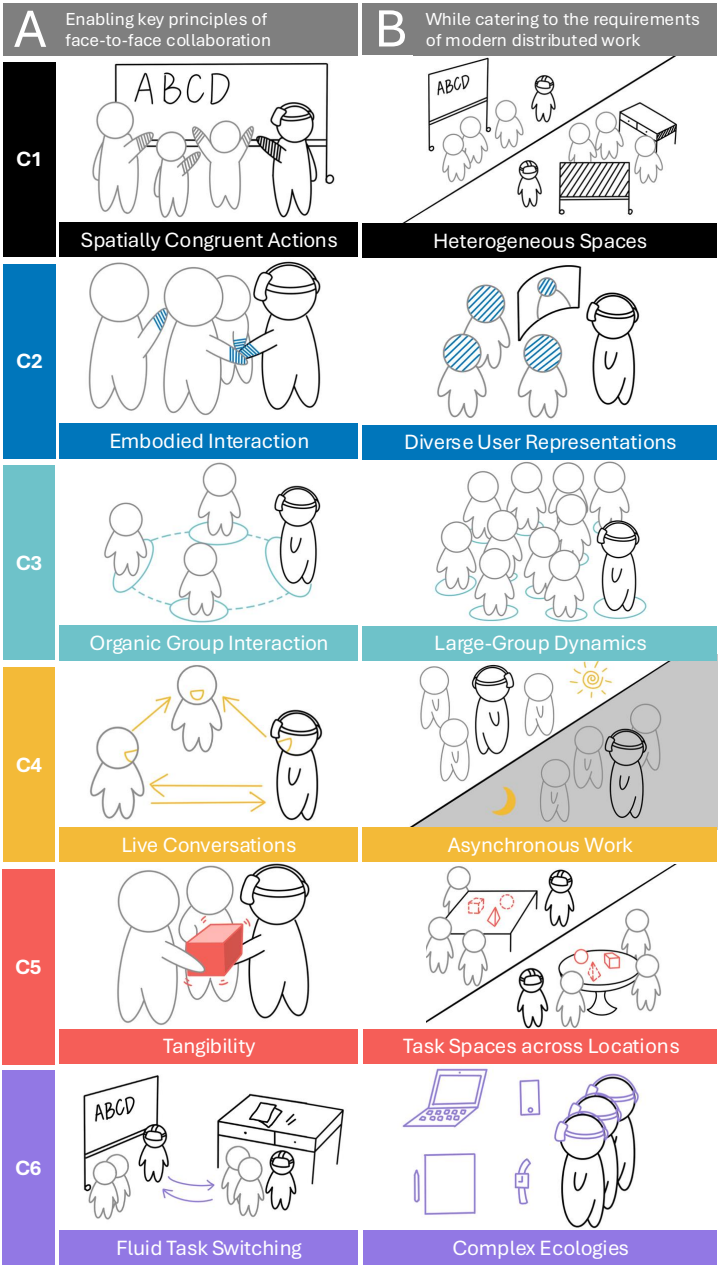


Figure 1.2: Based on identified tensions between (A) principles of face-to-face collaboration and (B) requirements of modern distributed work, this manuscript articulates six grand challenges, outlining a research agenda for distributed MR collaboration.

ern distributed work. It presents an organised overview of six grand challenges of distributed HMD-based MR collaboration for small groups. More specifically:

- We present six grand challenges and illustrate their relevance and inherent tension with a user scenario.
- We identify state-of-the-art progress and open research questions related to each grand challenge.
- Finally, we propose new specific research contributions to address the open questions.

As illustrated in Figure 1.2, each grand challenge addresses the tension between supporting a face-to-face principle (A) and navigating a modern work requirement (B). While much prior work has addressed these two sides individually, the tensions between them have largely remained unexplored. Each challenge is further articulated as a statement, summarised in Figure 1.3. The challenge statements and their corresponding research questions are elaborated in the following chapters 4—9. Before unpacking these, we will first explain why the research community needs to solve these grand challenges (Section 1.2) and clarify the scope of the agenda (Section 1.3).

1.2 From vision to grand challenges

Human-Computer Interaction (HCI) has an opportunity to play a pivotal role in the design of MR systems enabling face-to-face interactions in distributed work settings. The vision is not new—among others, The Office of the Future (Raskar *et al.*, 1998) and Holoportation (Orts-Escolano *et al.*, 2016) present compelling cases for MR as the medium for enabling more spatial collaboration over distance. However, despite the vast amount of visionary sources for the future of MR and collaboration (Grønbæk *et al.*, 2026), we found that the HCI community lacks a research agenda that clearly articulates the relevant challenges and open questions for researchers to contribute to *realising this vision in a modern distributed work context*.

Vision

Enabling distributed users to collaborate as if they were face-to-face, while each user's interactions remain grounded in their physical reality

Grand Challenges**C1**

Every collaborator can perceive others' actions as spatially **congruent** with their environment regardless of the **heterogeneous** configurations of others' respective environments.

C2

Every collaborator is assured that their **user representations** accurately portray their intended **embodied interactions**.

C3

Each member of a group should be able to use their real body to **organically** approach and depart from smaller group conversations, while maintaining awareness of the **larger group** around them.

C4

Each collaborator should be able to engage in **live embodied conversations** around task objects **across time**, allowing them to both review and contribute to past conversations for future decision-making.

C5

Every collaborator can communicate through spontaneous, symmetric interactions with **tangible objects**, with seamless replication and synchronisation of object state **across locations**.

C6

The group can **fluidly** transition between collaborative tasks, navigating the **complex** ecology of task items to spontaneously share contents both within MR and between MR and other modalities.

Figure 1.3: The vision and the six grand challenges for distributed MR collaboration, C1—C6, aimed to reconcile tensions between principles of face-to-face collaboration and requirements of modern distributed work.

Our manuscript articulates six statements (Figure 1.3) focused on the tensions between the vision of “being there” and modern-day work requirements. Starting top-down from these tensions, we unpack key challenges and open research questions to guide future research. The challenges were derived and refined based on two expert workshops; one formative workshop among the co-authors of the manuscript, and a follow-up workshop with 26 participants from the HCI community at the ACM CHI 2025 conference (Genay *et al.*, 2025) which helped us further expand on the challenges.

Prior work has aimed to address this gap through a *bottom-up* approach, either from systematic reviews (Ens *et al.*, 2019; Schäfer *et al.*, 2022; Sereno *et al.*, 2022) or expert workshops in the community (e.g., Mueller *et al.* (2020), Elvitigala *et al.* (2024), and Ens *et al.* (2021)). The bottom-up approach excels at providing an *overview* of a field, extrapolating trends from past efforts to guide future directions. However, in contrast, a vision-driven approach works *top-down*, starting from the vision. This can help the field *focus* its efforts and bring attention to overlooked directions, particularly where the focus is on work yet to be done rather than previous work. Our approach thus has the advantage that it can set a more future-oriented research agenda compared to those purely built from prior work.

Ens *et al.*’s recent survey on Collaborative MR (Ens *et al.*, 2019) is a seminal example of a bottom-up contribution that outlines past and future directions for the field. They highlight that research trends are emerging around (1) supporting more complex collaboration structures in time, space, and symmetry, (2) convergence and transitional interfaces, (3) empathic collaboration, (4) collaboration beyond the physical limits, and (5) social and ethical implications. However, what should guide the next endeavours on these trends? According to Mazzucato’s framework (Mazzucato *et al.*, 2018), solving societal challenges requires defining agendas with grand challenges within a concrete scope to guide the direction for new research projects. Our manuscript defines such an agenda building on the first trend identified by Ens *et al.* (2019): *supporting more complex collaboration structures in time, space, and symmetry*. Yet, in contrast to simply outlining a trend, our agenda outlines challenges driven by the *tensions* between these modern-day

collaboration complexities and the aim for MR to create a sense of “being there”.

1.3 Scope and position to related agendas

In our agenda, we specifically consider teams of 2 to 4 individuals; shown to be a common and desirable group configuration (see example by Kooloos *et al.* (2011)). Additionally, we limit our exploration to MR systems and scenarios for Head-Mounted Displays (HMDs). While we do not argue for or against the use of other MR displays (e.g., projection-based or handheld), we consider HMDs to currently strike the right balance for early adoption in modern teamwork, with being a highly immersive (taking over the user’s visual perception) and an accessible form of MR (cheap commodity devices) that does not require extensive room calibration or specialised setups, such as in the implementation of the Office of the Future (Raskar *et al.*, 1998).

Despite anchoring our exploration around HMDs, we have phrased the interaction challenges to be technology agnostic, so as to generalise to MR research beyond HMDs. We use the term Mixed Reality (MR) as the technological scope for the challenges. While MR has several working definitions (Speicher *et al.*, 2019), we deliberately set aside fully immersive systems in favour of a focus on physical reality (see vision statement in Figure 1.3). Therefore, our scope excludes VR as it typically bears no connection to the user’s real environment. Our agenda is concerned with scenarios in which the distributed team’s **physical spaces bear relevance to the collaborative activity**. The term Augmented Reality (AR) is too restrictive, as it implies a full visual perception of physical reality, thus MR is more aligned with our approach, as we aim to include solutions along the reality–virtuality continuum (Milgram and Kishino, 1994) that blur these boundaries, such as mixed-presence groupware (Krug *et al.*, 2026) and cross-reality and hybrid user interfaces (Hubenschmid *et al.*, 2026).

This agenda differs from most VR (and broader MR) visions. For instance, McVeigh-Schultz and Isbister (2022) argue that, instead of simply replicating physical face-to-face interactions, we should look for ways to provide social superpowers that go “beyond being there” (in the

words of Hollan and Stornetta (1992)). Examples of such features include enabling representations that remain “open to interpretation” (Sengers and Gaver, 2006), visualising physiological signals (Sasikumar *et al.*, 2022), and transforming interpersonal space (Bailenson *et al.*, 2004). This is not incompatible with MR, and, as Billinghamurst and Kato put it, distributed MR solutions can support both seamless face-to-face interaction *and* enable collaborators to go beyond being there (Billinghurst and Kato, 1999a).

While we agree with both of these propositions (they do lead to exciting new possibilities, e.g., Piumsomboon *et al.* (2018), Piumsomboon *et al.* (2019), and Grønbæk *et al.* (2024)), we also contend that MR offers a more succinct value proposition—i.e., to bring face-to-face principles back into distributed team collaboration (see Figure 1.2). Mazzucato’s framework suggests that in order to solve a larger societal challenge, we must define multiple agendas that work in *parallel* (Mazzucato *et al.*, 2018); i.e., a well-scoped agenda should not hold back the advancement of other agendas. Through this lens, we emphasise that working to create a sense of “being there” does not hold back other agendas to harness the unique affordances of MR. In fact, we do not argue that a collaborative experience that feels like face-to-face is the right solution for every situation, but rather that it is an often desirable yet unsupported mode of interaction in modern distributed teamwork.

2

Background

We will now unpack the argument that there is a mismatch between human face-to-face interaction skills and characteristics of modern distributed work. To provide the basis for this argument, we draw inspiration from socio-spatial theories and CSCW to highlight important principles of face-to-face interaction (2.1), and identify requirements of modern distributed work that lie in tension with these (2.2), as illustrated by the overview in Figure 1.2.

2.1 Key principles of face-to-face collaboration

Understanding the principles integral to human communication skills can explain why face-to-face meetings are often more effective and enjoyable than online interactions. More importantly, it offers a clear vocabulary for examining how MR can help mitigate the limitations of current distributed communication tools.

To begin, one of the main characteristics of face-to-face communication is that it is inherently spatial. Humans heavily rely on social context and spatial information to interpret other people’s mental states and course of action, e.g., by observing walking directions, body posture, orientation, and eye gaze (Hall, 1966; Lawson, 2007). When we share the

same physical space, we can interpret **spatially congruent actions** as all present collaborators share the same spatial frame of reference. This shared reference can be used to communicate intentions, bring attention to something, and support situational awareness (Hall, 1966; Knapp and Hall, 2009; Lawson, 2007). The CSCW literature highlights the need for any groupware system to support such awareness (Gutwin and Greenberg, 1996). Socio-spatial theories have been developed to describe how this awareness is inherent in co-located collaborative settings, understanding how people use space to interact with each other (proxemics (Hall, 1966; Grønbaek *et al.*, 2020; Marquardt *et al.*, 2012)), how verbal and body language direct attention within space (deixis (Mentis *et al.*, 2012; Kendon, 1997)), how space is partitioned and negotiated (personal space and territoriality (Lawson, 2007; Sommer, 1969; Tuddenham and Robinson, 2009; Scott *et al.*, 2004)), and how collaborators switch between individual and group activities (coupling styles (Tang *et al.*, 2006)).

Closely linked to spatiality is how face-to-face settings facilitate **embodied interactions** (Dourish, 2001; Harrison and Dourish, 1996). Our body not only allows us to move through space but also serves as a way to communicate non-verbally. Subtle bodily movements, such as a slight body lean, a shared glance, or a facial expression, are powerful signals and cues that help us establish common grounding more efficiently than with words alone (Kendon, 1990; Knapp and Hall, 2009; Hall, 1966; Kendon, 1997).

When face-to-face, we also use space to facilitate **organic group interactions**, i.e. fluidly emerging without pre-defined structure (Kukshinov *et al.*, 2024; Li *et al.*, 2021). Such interactions become more complex as the group size increases and individuals must simultaneously interpret the actions of multiple interlocutors. Forming subgroups can help structure conversations and change collaboration scales to more manageable sizes (Lawson, 2007; Wong *et al.*, 2024a). To facilitate the creation of subgroups, people rely on common socio-spatial rules (Krogh *et al.*, 2017; Kendon, 1990; Tong *et al.*, 2016) and often spontaneously arrange themselves in facing formation patterns, formalised as the f-formation system (Kendon, 1990). These patterns further arise around physical objects and other spatial features, which serve as anchors for

group formation (Marshall *et al.*, 2011).

Of course, another primary reason people often meet face-to-face is to engage in synchronous, **live conversations**. Receiving direct feedback is essential to advance many tasks, whether it be solving problems, planning, or making decisions. Real-time conversations allow collaborators to negotiate and converge on solutions much more rapidly than with asynchronous conversation (e.g., via e-mails), fostering real-time feedback, rapport, and empathy (Boehm, 2009).

The importance of sharing the same space for collaboration extends beyond the immediacy of live interactions to providing **tangible interactions** (Stout and Chaminade, 2012). Many collaborative tasks (such as design work, assembly, or demonstration) require working with physical objects and manipulating them together. Meeting face-to-face enables sharing equal views and access to these objects without further ado. In addition, when collaborating, people frequently incorporate physical props and tools into their interactions (Ishii and Ullmer, 1997; Dourish, 2001) as a form of cognitive scaffolding (Clark, 1996), thereby anchoring attention and reducing cognitive load (Kirsh, 2010).

Lastly, face-to-face settings facilitate coordination and **fluid task switching** by enabling groups to organise tools and work materials in a shared workspace (Perry and O'Hara, 2003; Mentis *et al.*, 2012; Klemmer *et al.*, 2006). Their arrangement and display in space support embodied and spatial cognition: with a simple gesture, or by moving between objects, collaborators can fluidly shift their collective attention from one artefact to another, whether physical or digital (Mentis *et al.*, 2012).

The above principles of face-to-face interaction are deeply tied to our ability to collaborate effectively. However, our key point in this manuscript is that these six particular principles are significantly challenged by the requirements of modern distributed work.

2.2 Requirements of modern distributed work

The nature of work has undergone significant changes in recent years, with workspaces becoming more varied and working times more flexible, and as a consequence, collaborative work is increasingly reliant on digital

tools. While this shift offers many advantages, it often comes at the expense of the face-to-face collaboration principles highlighted in the previous section.

First, collaboration over large distances has become commonplace: remote work, once the exception, was normalised by the COVID-19 pandemic, and modern teams often collaborate across **heterogeneous spaces** that are not necessarily designed for work (e.g., homes, cafes, and office spaces, with varying layouts). Therefore, teams increasingly rely on digital tools like video conferencing to meet. While collaborators gain flexibility with such tools (Collins *et al.*, 2022), they lose the benefits from previously outlined spatial aspects of communication.

Another consequence of distributed work is that it becomes impossible to *directly* see one's collaborators. Providing real-time visuals of them and what they are doing requires digital **user representations**. In videoconferencing, this is realised as a camera feed on a 2D screen, usually displaying only their upper body and constraining their movements to stay within the camera's field of view. Naturally, this makes it hard to observe embodied interactions and cues that facilitate efficient communication. The user's uncertainty about how they are perceived through the mediating technology can lead users to continuously monitor their self-view and invest further efforts, such as to ensure good lighting and perspective, or use video filters. This has a consequence on their cognitive load, known as "Zoom fatigue". Bailenson's explanations 2021 for this fatigue include (1) the cognitive efforts of monitoring "translated" or reduced visual access to nonverbal gestures, (2) having to exaggerate one's nonverbal gestures to make them visible through video, (3) the imposed immobility or lack of embodied movement caused by having to sit still in front of a camera, or (4) the fatigue from inconsistent proxemics with eye gaze at a close distance or staring at your own image.

On the flip side, the flexibility offered by digital tools has made it possible to collaborate with a larger number of people simultaneously, enabling **group interactions at scale**. Zoom now accommodates up to 1,000 participants and can host large events like conferences without the costs of hiring a venue and travelling there. While this opens up new opportunities for how events and group work can be organised, current

tools still fail to support fluid group dynamics. For example, break-out rooms that were proposed to enable parallel discussions end up isolating groups from each other and removing the spontaneity of walking up to someone. Solutions like GatherTown (Gather Presence, Inc., 2023) offer more embodied interactions by letting users navigate a virtual space using an avatar that activates webcams for video communication as users get close to each other. However, the user expression remains limited by the lack of embodied interaction with the 2D avatar.

In addition to large distances, modern work is also increasingly happening **asynchronously**. This established practice is highlighted in shift work, where workers collaborate within the same physical environment within designated periods throughout the day. However, daily exchanges between teams located in entirely different countries have now also become possible and greatly facilitate the internationalisation of collaboration. Thereby, collaborators can no longer rely on the benefits of shared space and time, facing the difficulties of finding time slots during working hours to meet potentially across various time zones. Yet, some tasks remain better performed live, such as decision-making, negotiation, and dynamic interactions that require immediate feedback (Baym *et al.*, 2021), and this has sparked the development of new asynchronous collaboration tools such as SlackClips¹ and Loom² to provide more expressive communication through video recordings, effectively creating an asynchronous alternative to video conferencing. However, such solutions suffer from the same issues of synchronous video calls described before and potentially amplify these due to the lack of immediacy.

As teams become more and more distributed, the tasks they do also undergo changes in how they are executed. Rather than centralising work within one location, modern work often happens in liaison over multiple **task spaces across locations**. This allows realising tasks at their most relevant locations while keeping access to remote expert knowledge from anywhere. However, tasks that depend on physical objects, such as prototypes, may suffer with this distributed operating

¹<https://slack.com/features/clips>

²<https://www.loom.com/>

mode, as remote collaborators who do not have access to the physical object often have little opportunity to participate. Frequently, local participants will dominate the collaboration, sometimes creating feelings of exclusion—an experience known as *primary room dominance* (Karis *et al.*, 2016; Saatçi *et al.*, 2020; Grønbæk *et al.*, 2021). This dominance is especially accentuated in hybrid settings, involving both co-located and remote participants (Neumayr *et al.*, 2021; Bjørn *et al.*, 2024; Saatçi *et al.*, 2020; Neumayr *et al.*, 2018). To remedy this, MR research has often proposed asymmetric systems where the remote users are “brought” into the local user’s physical space using VR. Although this is an effective solution, it sacrifices the remote user’s vision of their own physical space (Irlitti *et al.*, 2023; Fidalgo *et al.*, 2023b; Faridan *et al.*, 2023).

Lastly, modern work has been experiencing a rapid multiplication of digital tools and devices to cover a broad range of tasks from home. Numerous improvements have been made to existing collaborative software, resulting in a smoother workflow when juggling tasks by connecting tools together through plugins and APIs or integrating them into super-systems (e.g., Microsoft 365). Efforts were made to design dedicated meeting spaces such as Cisco TelePresence rooms (O’hara *et al.*, 2011)), but simpler collaboration tools like Zoom remained the market leaders, providing more frictionless interaction fit for modern work. Nevertheless, the modern knowledge worker still needs to adapt to the ever-increasing number and **complex ecologies** of tools. Not only must they navigate a large ecosystem of software and devices (Bødker and Klokmoose, 2012), but they must also alternate between types of computer-supported collaborative work paradigms, e.g., switching from co-located to remote and from synchronous to asynchronous collaboration (Johansen, 1988).

As exemplified above, the gains in broadening collaborative participation through modern digital tools have often been paid for by sacrifices in the core elements of face-to-face collaboration. It is this tension that forms the bedrock of our agenda.

3

Scenario: a distributed MR workshop

Throughout the following chapters, we will dive deep into the six grand challenges. To anchor these MR interaction challenges in the modern requirements of distributed team collaboration, we describe a concrete usage scenario: a distributed MR workshop.

Prior research often refers to collaborative creative tasks, such as architectural design, to demonstrate a use case where distributed collaboration benefits from face-to-face principles such as symmetric tangible interaction and embodied communication (e.g., Underkoffler and Ishii (1999), Broll *et al.* (2004), Ishii *et al.* (2004), Piper *et al.* (2002), Verlinden *et al.* (2003), Rauterberg *et al.* (1998), and Grønbæk *et al.* (2024)). We base our grand challenges on this type of scenario. Specifically, we unpack a distributed workshop scenario with a group of architectural designers (see Figure 3.1).

We acknowledge that requiring symmetry and full embodiment is not desirable for all types of collaboration. In fact, many creative problem-solving tasks can potentially be solved without it. Historically, collaborative MR research has mostly focused on such cases: either physically co-located collaboration where symmetry and embodiment come for free (e.g., Szalavári *et al.* (1998)) or asymmetric distributed col-

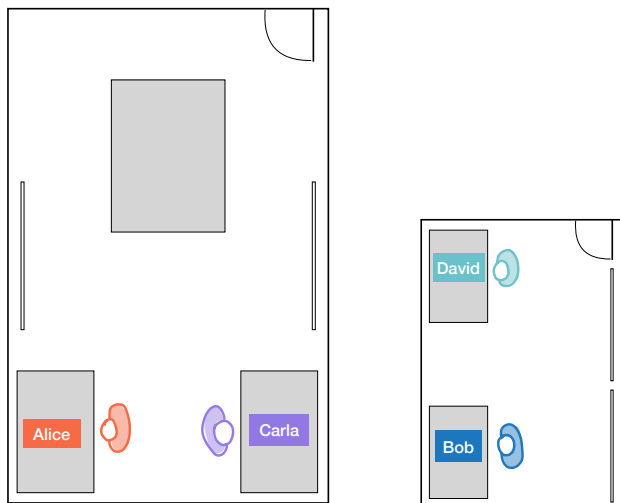


Figure 3.1: *A distributed team scenario*: Alice, Bob, Carla, and David are architects undertaking a distributed meeting enabled through MR. Their goal is to put the final touches on their presentation of a building concept in preparation for an upcoming workshop with stakeholders. Alice and Carla are co-located in the company headquarters, while Bob and David are co-located in a branch at another location. The distributed team works across heterogeneous spaces, with variable group sizes and frequent task switching with both physical and digital objects, and they work across different time zones with minimal overlap in work hours.

laboration where remote assistance is provided asymmetrically through partial embodiment (Fidalgo *et al.*, 2023b; Huang *et al.*, 2022; Kim *et al.*, 2020). By focusing on a single physical environment, these scenarios are comparatively simpler than the one we explore: it is more challenging to support distributed collaborators to stay grounded in their respective local spaces while perceiving remote collaborators join them in their space. At the same time, if we could enable experiences with a high degree of symmetry and embodiment—even under the challenging conditions of distributed work—we could potentially reduce the need for travelling, with a major impact on making creative collaboration across

distance more sustainable.

In the following sections (4—9), we outline six grand challenges **C1—C6** aiming to provide such an experience.

4

C1: Locally congruent actions from remote heterogeneous spaces

MR offers the unique potential to reintroduce our **spatial** face-to-face skills back into distributed collaborations. In contrast to VR, MR can anchor interaction in local physical space, thereby allowing users to move around the floor using their real bodies and engage with physical objects and furniture in their vicinity. However, as outlined before, this property is in tension with the real-world circumstance that distributed work might happen across highly diverse physical environments. This goal can be expressed as follows:

Grand Challenge C1: *Every collaborator can perceive others' actions as **spatially congruent** with their local environment, regardless of others being in **heterogeneous spaces**.*

4.1 Research challenge

To realise this goal, we must solve the challenge of congruency, i.e., to establish a congruent MR interaction space despite differences in the layout of the distributed users' heterogeneous

environments. Heterogeneities may consist of different room shapes and sizes or dissimilar furniture layouts and characteristics. To illustrate the challenge, consider the scenario illustrated in Figure 4.1.

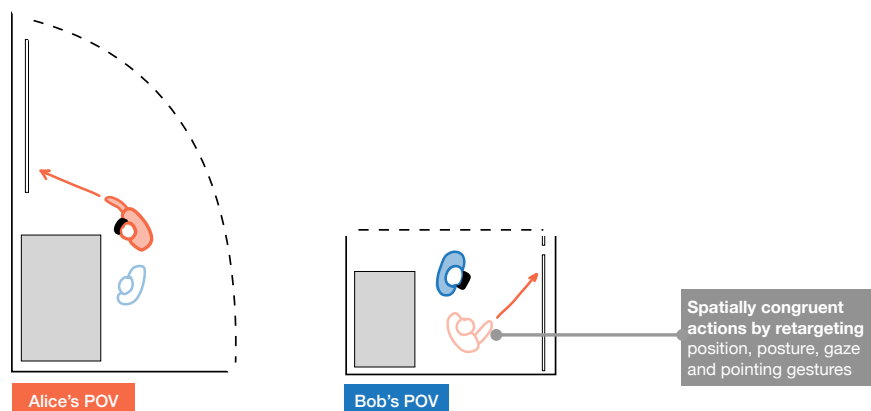


Figure 4.1: *Spatial congruency challenge*: The architects break out into pairs. Alice wants to collaborate with Bob across their two sites, using her desk and whiteboard. Bob also has a desk and a whiteboard in his office. Meeting in MR, their respective whiteboards are *blended* into one, and so are the desks. They can draw together on their blended whiteboards and point at sketches while virtually standing next to each other. They can also refer spatially to virtual documents spread out on their blended desks.

In this scenario, if Alice walks from her whiteboard to her desk while pointing at something on her whiteboard, this movement needs to be translated congruently in Bob's space, in relation to his own desk and whiteboard, regardless of their different relative positioning. To make these interactions appear congruent, the challenge then becomes to reconstruct the user's actions in remote space, so that they can consistently refer to objects and physical features in any remote user's context. Translating such actions is difficult given there are many nuances to non-verbal communication, but research has started to identify strategies.

4.2 State-of-the-art research and open questions

Understanding how to achieve congruency when distributed spaces are heterogeneous is an active area of research. Sharing **physical features** of space, such as furniture, desks, whiteboards, and room boundaries, was shown to be essential for collaboration because it provides a congruent reference space for deixis and movement (Yoon *et al.*, 2022; Jo *et al.*, 2015; Lehment *et al.*, 2014; Fink *et al.*, 2022; Herskovitz *et al.*, 2022; Grønbæk *et al.*, 2023; Sra *et al.*, 2018; Wang *et al.*, 2022; Yoon *et al.*, 2021; Sousa *et al.*, 2019; Johnson *et al.*, 2023; Johnson *et al.*, 2021). In MR, these physical features can be harnessed to create functional areas that are meaningful in a workspace blended across locations. For example, “let’s go to the whiteboard” can transparently refer to every collaborator’s physical whiteboard as if there was only one.

The mapping of these features across heterogeneous spaces provides a stable context for avatars to be represented, ensuring that their actions are grounded in any user’s physical reality. One intuitive approach for this mapping is to physically align (or homogenise) the distributed spaces by moving around or changing the furniture so that the layouts are identical or very similar. Famous examples are *Room2Room* (Pejsa *et al.*, 2016) and *Holoportation* (Orts-Escolano *et al.*, 2016). However, this physical alignment approach is not feasible in most cases.

Recent explorations have proposed alternative approaches (Grønbæk *et al.*, 2023; Wong *et al.*, 2024b), distinguished by whether they attempt to provide global or partial alignment. Partial alignment blends only a part of the distributed environments, leaving the rest misaligned (e.g., Fink *et al.* (2022), Grønbæk *et al.* (2023), Herskovitz *et al.* (2022), Johnson *et al.* (2023), Johnson *et al.* (2021), and Thoravi Kumaravel *et al.* (2019)). This means that movement is only congruent within subspaces of the larger workspace, and transitions must be implemented to convey the realignment (Fink *et al.*, 2022; Grønbæk *et al.*, 2023). On the other hand, global alignment applies spatial transformations to blend the whole distributed environments, regardless of their mismatch (e.g., Yoon *et al.* (2022), Jo *et al.* (2015), and Wang *et al.* (2022)). This often leads to distortions that cause a temporary inability to produce congruent action cues (Yoon *et al.*, 2022).

These solutions present different design trade-offs when it comes to enabling congruency of avatar deictic and proxemic cues, i.e., how avatars reference spatial features, move around, and position themselves relative to each other and the shared space. Head and gaze orientation are important for us to discern information such as “Is my collaborator looking at the whiteboard?”. To support **deictic cues**, previous work has proposed redirecting spatial references through pointing gestures and congruent attentional cues (Fidalgo *et al.*, 2023a; Piumsomboon *et al.*, 2018; Yoon *et al.*, 2021; Sousa *et al.*, 2019). For example, Piumsomboon *et al.* (2018) calculate the relative positions of local and remote objects to dynamically transform pointing vectors and align avatar gesture directions meaningfully with local space.

Supporting congruent **interpersonal distance and orientation** relative to each user’s local space is also important for us to rely on proxemics, i.e., our innate sensitivity to the spatial nature of social interaction. As long as a system can provide the perception of a consistent and stable interpersonal distance and orientation between collaborators’ avatars, congruency is in place. Solutions for this have notably been proposed by Ballendat *et al.* (2010), using data structures to store information like distance, angle, and proxemic areas and adapt the spatialisation of virtual content. However, proxemic behaviours of avatars may be distorted across spaces, and how to appropriately configure them remains a challenge.

To inspire future research on achieving spatial congruency under distributed work conditions, we outline key questions raised by this challenge (labelled boxes indicate the **type of contribution** to HCI, following Wobbrock and Kientz (2016)).

- **System** **Toward scalable solutions:** Prior prototype solutions to congruency focus on the simple case of two remote users across two spaces. Echoing Ens *et al.*’s argument 2019, it is clear that congruency challenges are further exacerbated when more than two heterogeneous spaces must be aligned. While a few works have contributed to this concern (Ens *et al.*, 2019; Wong *et al.*, 2024a), there is a lack of attempts to prototype solutions in this direction. *How can systems scale to more than two heterogeneous spaces with*

more than two users in remote or hybrid team configurations?

- Empirical and Methodological Longitudinal studies of how MR affects the notion of place:** Empirical work tends to focus on technical performance, usability or design, with no longitudinal evaluation. However, with teams' adoption of solutions that map distributed spaces, we should investigate what this means for the notion of "place" (i.e. the cultural meaning of the space). Harrison and Dourish 1996 suggest the sense of place is shaped by everyday practices. Within blended spaces, if each person's space is different, then their sense of place may also differ from each other. While recent work (Wong *et al.*, 2024a; Wong *et al.*, 2024b) has started to ask these questions from a conceptual perspective, there is an opportunity for future research to make both empirical and methodological contributions by investigating how teams may adopt MR for distributed meetings, with longitudinal accounts of regular MR usage across heterogeneous distributed workplaces. This approach could address the question: *How can collaborative MR support "place-making" in distributed settings?*
- Survey and Theory A systematic taxonomy of heterogeneous spaces:** As mentioned earlier, enabling congruent deixis and proxemics requires navigating design trade-offs. However, we must develop a clear vocabulary and overview of the factors that affect these tradeoffs to guide future research. More conceptual work is needed to understand how to design collaborative MR for different task and space contexts. The right solution certainly depends on the physical environment's relevance to the task at hand. While recent work mapped out a collaborative task taxonomy for MR (Ghamandi *et al.*, 2023) and proxemic dimensions of collaboration across heterogeneous spaces (Wong *et al.*, 2025), these theoretical frameworks must be further tested and expanded to the wealth of published solutions in the context of heterogeneous spaces. Future work should aim to develop taxonomies that help researchers navigate: *What are the assumptions that each potential solution makes when it comes to supporting spatial collaboration*

across heterogeneous spaces?

5

C2: Embodied interactions through diverse user representations

MR has the potential to mitigate the ‘all-day mirror’ fatigue of video-conferencing (Bailenson, 2021) by giving all users embodied representations in the form of avatars. Each avatar can exhibit **literal embodied interactions**, allowing for the direct coupling of local user behaviour and remote virtual representation. A fundamental strength of MR over video-conferencing technologies is that “harnessing full face and body motion capture can make social interaction [...] very similar to face-to-face interaction” (Rogers *et al.*, 2022): each user can freely move their avatar body around to change perspective during a distributed collaboration, to point and gesture (Dodds *et al.*, 2011), and avatars can even mimic their live facial expressions (Zhang *et al.*, 2024). Moreover, MR enables a stronger kinesthetic experience compared to many collaborative VR applications by constraining the interaction space such that users can, for example, physically walk to access places. Beyond these spatial and kinesthetic aspects, a virtual user representation can offer users an opportunity to define their appearance. Modern avatar personalisation approaches permit users to generate humanoid avatars based on scans of themselves, and customisation tools e.g., Meta Avatars

SDK¹, allow the manual configuration of avatar body dimensions, colour of skin, eyes and hair, clothing and even accessories, even beyond the realistic.

However, as highlighted in C1, remote user representations can be distorted in remote space. This may be caused by the spatial alignment, but also by a varying fidelity of body appearance and tracking (Genay *et al.*, 2021; Bonfert *et al.*, 2025), whereby facial and bodily expressions may not be faithfully communicated (e.g., crossing legs, frowning, waving hands). This decoupling raises additional challenges when we consider the level of subtlety that some nonverbal cues can have. Therefore, when the above distortions may occur, the embodied communication of intent becomes especially difficult, as captured in the following grand challenge:

Grand Challenge C2: *Every collaborator can express their **intent** through embodied interactions while being portrayed by diverse user representations.*

5.1 Research challenge

To take steps towards achieving this goal, we need solutions that **provide remote users with virtual representations that best support embodied actions within each user’s local space**. In mediated spaces, users have two representations: one physical (their real bodies) and one virtual (their digital user representation). As such, the question becomes how faithful one’s remote representation should be to enable effective collaboration in blended spaces. The scenario illustrated in Figure 5.1 raises the significance of this challenge in hybrid settings, where collaboration is partially distributed.

In this scenario, Alice and Carla both see each other’s real bodies, letting them communicate through subtle non-verbal cues. For example, as Alice sees Carla shifting her posture to show she’s about to stand

¹<https://developers.meta.com/horizon/documentation/unity/meta-avatars-overview/>

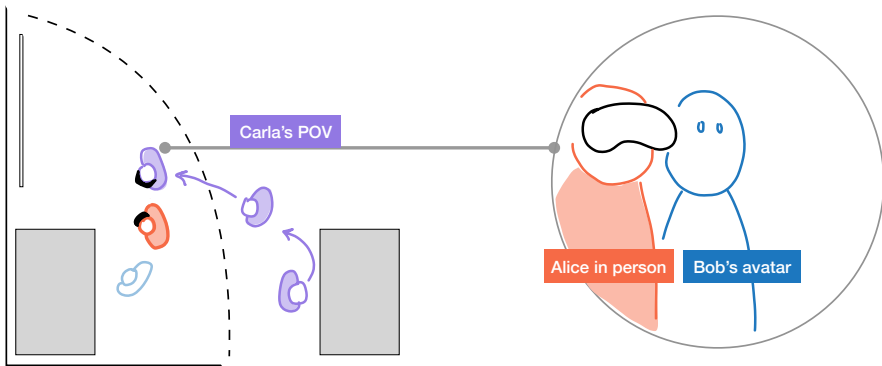


Figure 5.1: *User representation challenge*: Carla is physically co-located with Alice, working at her desk. As Alice starts an MR meeting with Bob to prepare for the upcoming workshop, Carla puts on her headset and enters Alice’s zone of interaction. She can now perceive Bob appearing as a virtual 3D avatar, standing next to Alice. On Bob’s side, Carla and Alice are also represented as avatars that transparently communicate their non-verbal cues. They all see each other’s faces as unoccluded, although they are all wearing headsets that hide their real faces in their local spaces.

up but still needs to finish writing something, she knows Carla has not forgotten their meeting with Bob and that she will join in an instant. However, as Alice wears a headset, Carla does not have a clear, direct view of her face. Bob, on the other hand, can only express what his tracking technology and virtual representation afford communicating. He might be unsure whether Alice and Carla properly see and understand his gestures, such as when he raises his eyebrows to communicate surprise, as we commonly lack self-views (e.g., virtual mirrors) for monitoring our avatars (Krug *et al.*, 2025). He might even deliberately exaggerate his facial expression, in doubt (Kukshinov *et al.*, 2024). Further, he may have chosen a virtual avatar that does not closely resemble his own appearance, but instead represents him as his favourite animal (e.g., a panda). While this brings a humorous note to the task, it may not be appropriate for a professional work context, may make facial expressions harder to read, and Bob may not even be immediately

recognised by Alice and Carla. Vice versa, Bob also perceives Alice and Carla as virtual user representations, which may be similarly distorted.

This scenario illustrates the diverse aspects of expressivity that must be considered when representing people in MR. While the development of high-fidelity rendering and scanning technologies is out of our scope, it falls on HCI to design appropriate user representations for remote collaboration. The challenge resides in identifying the traits that are necessary for satisfactory self-expression, while avoiding spending cognitive effort thinking about how they are represented in remote locations.

5.2 State-of-the-art research and open questions

There has been significant research exploring the opportunities for freedom of self-representation and identity through avatar ownership in virtual environments (Kolesnichenko *et al.*, 2019; Jang *et al.*, 2024). For example, virtual user representations may offer the opportunity to appear as idealised versions of ourselves (Bujić *et al.*, 2025), express sentiments, make statements, or take a new perspective by embodying someone else (Banakou *et al.*, 2013; Banakou *et al.*, 2016; Seinfeld *et al.*, 2018), or perhaps to mask or highlight particular disabilities to manage expectations (Zhang *et al.*, 2023; Gualano *et al.*, 2024). However, when considering modern collaborative work, a faithful representation is often desired to support consequential communication and increased productivity (Gutwin and Greenberg, 2004; Bonfert *et al.*, 2023; Do *et al.*, 2024), where customization options may serve to further adapt a representation to a specific scenario (e.g., a professional look for work or a casual dress for entertainment) (Kiltene *et al.*, 2013), comparable to how we might pick clothes for an occasion in the real world. Importantly, to have effective workspace awareness in distributed MR collaboration, remote user representations should convey an unambiguous sense of *who* each user is (Gutwin and Greenberg, 1996; Steed and Schroeder, 2015) (e.g., humanoid form, a face free from occlusion). In the example scenario, when Carla enters the meeting space, Bob's experience must be in a similar vein to if he were co-located with Alice and Carla; he should instantly comprehend that an additional person has joined the meeting, and that this person is Carla.

In recent years, there have been notable technical achievements in producing highly realistic reconstructions of a user’s appearance (Shen *et al.*, 2023; Dou *et al.*, 2016; Orts-Escolano *et al.*, 2016; Waltemate *et al.*, 2018; Salagean *et al.*, 2023), as well as tools for avatar customisation (Morris *et al.*, 2023; Gonzalez-Franco *et al.*, 2024; Ouhnni *et al.*, 2025). Further, researchers have worked towards understanding the impact of user representation appearance on social behaviour (Ma *et al.*, 2021; Kolesnichenko *et al.*, 2019; Shen *et al.*, 2023; Michael *et al.*, 2017; Latoschik *et al.*, 2017; Kukshinov *et al.*, 2024; Fink *et al.*, 2024; Yee and Bailenson, 2007; Aseeri and Interrante, 2021; Yoon *et al.*, 2019). Moreover, MR research has explored how to facilitate unoccluded views of a user’s non-verbal facial expressions when wearing head-mounted displays through avatar representations (Orts-Escolano *et al.*, 2016; Ma *et al.*, 2021; Kullmann *et al.*, 2025). In the case of embodied interactions, recent approaches have enabled reconfigurable meeting spaces with high-resolution camera arrays, capturing every user and object within the space (Irlitti *et al.*, 2024; Maimone and Fuchs, 2011; Orts-Escolano *et al.*, 2016). This process allows for a transition from a local physical environment into a *hybrid blended space*, allowing for the seamless representation of both local and remote objects and people (Irlitti *et al.*, 2024). For the individual, this experience is also unnoticeable, as their remote avatar is immediately instantiated and mapped directly to their body pose.

As discussed in C1, the heterogeneity of spaces may necessitate a decoupling of embodied user inputs with avatar outputs. However, the boundaries of when and how to adapt avatars are unclear, such as when to go beyond locational-based transformations (e.g., pointing, head-turning, re-positioning, resizing) (Sousa *et al.*, 2019; Wang *et al.*, 2024; Piumsombon *et al.*, 2018; Watkins *et al.*, 2024), or even going so far as to modify an avatar’s underlying embodied behaviour (i.e., modifying eye gaze (Roth *et al.*, 2018; Bailenson *et al.*, 2004), enhancing or dampening expressive reactions (Thomas *et al.*, 2022; Oh *et al.*, 2016; Pandita *et al.*, 2020), or changing cultural behaviours (Hasler and Friedman, 2012)).

Finally, there are unique challenges when considering the partially distributed arrangement of hybrid teams (Neumayr *et al.*, 2018), and

the compromises that must be made to ensure congruency. Prior work has started to explore how to reduce these asymmetries (Gugenheimer *et al.*, 2017; Irlitti *et al.*, 2023), but social norms also play an important part in achieving an acceptable resolution to these tensions.

Taking stock of the above insights, the following questions must be addressed.

- **Empirical Understanding the implications of avatar distortions:** Such distortions can refer to movement remapping, or also deviation of the avatar from the user's actual appearance due to, e.g., low detection or transfer fidelity (Bonfert *et al.*, 2025) or intentional customisation. So far, the research has centred around technical proof-of-concept evaluations (e.g., Yoon *et al.* (2022)), with a lack of empirical work on how it affects collaborative work processes. Future work should address such questions. When avatar distortion is seamlessly executed, users within the local environment can use their bodies to convey their associated actions without any additional cognitive load. However, there are many implications to consider if distortion becomes the norm. *How do avatar distortions affect communication during collaborative tasks? To what degree can distortions be hidden, and when should they be transparent to users? Who is in control of these distortions (e.g., settings, customisations), and how do users recover from communication breakdowns caused by incorrect distortions?*
- **System Maintaining the integrity between multiple simultaneous representations of the same user:** Research on technical solutions to the challenge of hybrid (partially distributed) team configurations is sparse. In such configurations, representations of the same user will be simultaneously present across multiple locations — and some collaborators will be physically co-present with some while virtually co-present with others. This complexity increases with more users and more spaces. *How can we maintain consistent user appearances across different locations in distributed MR meetings to prevent confusion as collaboration scales up?*

6

C3: Organic group formation within large distributed teams

MR offers the opportunity to enable **organic** group formation and conversational floor management in distributed settings. By simply walking in their local physical space, users can easily switch from being in a group to standing in plenary formation or starting a one-on-one conversation. However, the challenge arises when accommodating groups within the constraints of their physical environments, such as when the group is too large for a given user's room. MR solutions must reconcile this tension, captured in the following grand challenge:

Grand Challenge C3: *Each member of a group should be able to use their real body to **organically** approach and depart from smaller group conversations while maintaining awareness of the **larger group** around them.*

6.1 Research challenge

To fulfil the goal of this grand challenge, we must address the problem of managing spatial group dynamics within physical environments of

limited sizes. Specifically, future MR systems **must accommodate fluid group formations of varying sizes and support a sense of interaction spontaneity, despite constraints in space.** Continuing our scenario, Figure 6.1 illustrates the challenge.

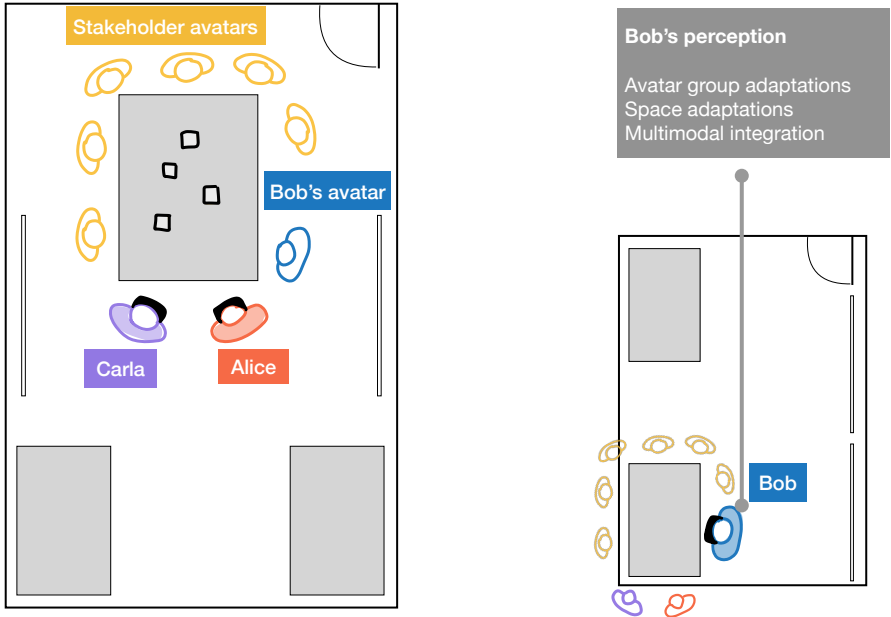


Figure 6.1: *Group challenge*: The day of the workshop has arrived, and the stakeholders are now in the architects’ workspace, waiting to hear about the concept they have been working on. The meeting begins with a plenary session, during which Alice gives a presentation. As Alice wraps up, she invites the attendants to join either Bob, Carla, or David and form small breakout groups in which they will lead more in-depth discussions. Each group steps away to engage in conversations without disturbing others. Meanwhile, Alice walks around the different groups to oversee that general progress is happening and sometimes spontaneously drops in to comment.

Enabling fluid group dynamics in MR presents a significant challenge (Baym *et al.*, 2021; Ens *et al.*, 2019). Indeed, the ability to mirror group formations is constrained by the heterogeneity of users’ spaces outlined

in C1 and by the ability to communicate subtle nonverbal cues detailed in C2. This challenge is amplified in scenarios requiring the fluid transition between various spatial formations integral to effective collaboration. Ens *et al.* (2019)’s review identifies a lack of work involving more than two users, leaving a gap in our understanding of how to enable these dynamics effectively.

6.2 State-of-the-art research and open questions

Social VR applications have brought large-group interactions a step closer to face-to-face settings. Compared to videoconferencing, users can embody avatars in the first person (Maloney *et al.*, 2021), offering the most advanced reproductions of spatial group dynamics. Williamson *et al.* (2021) and Williamson *et al.* (2022) demonstrated how VR allows mirroring physical proxemic behaviours and group formations. However, their study also shows that although responding to the necessity of fitting many users, large VR environments can hinder social cohesion and group interactions. This is reflected in other studies (Rasch *et al.*, 2023; Di Luca *et al.*, 2021; McVeigh-Schultz and Isbister, 2021) that found large virtual environments, which require teleportation to move around, tend to break the continuity needed for organic social navigation, ultimately complicating efforts to maintain a mental map of the group dynamics.

MR allows recentring group interactions around the participants’ local environments (Wong *et al.*, 2024a). In contrast to VR, this leverages the physical space to enable group formations according to the environment’s physical features. For example, while Alice and Carla are co-located with stakeholders in a workshop room, Bob joins remotely from his smaller home office and sees them together with him (Figure 6.1). However, what happens if there is not enough space to fit everyone on Bob’s side? MR offers unique opportunities to address these situations, but it also requires making important trade-offs: if no solution is available, some participants ultimately may have to appear at impossible locations, making them physically inaccessible and breaking the illusion of plausibility (Slater *et al.*, 2022).

This scenario describes two user needs: the MR system (1) must

support fitting large groups of users into physical spaces that are potentially too small to accommodate full-scale embodiment of all group members at once, and (2) it must provide equal means of interaction for both local and remote user representations. Although few works have explored solutions to this problem, a few offer initial ideas and insights into the challenges.

In a study with group facilitation experts, Wong *et al.* (2024a) explored the design space of organic group arrangements in distributed MR collaboration, identifying a set of design patterns. A selection of these patterns particularly addresses **how to meaningfully arrange large groups of avatars in confined spaces**. First, they propose to define zones in the user's local environment that represent functional areas within a larger shared workshop space (e.g., the local whiteboard area serves as the plenary zone, whereas the desk serves as the break-out group zone). Second, they suggest that avatar representations can be dynamic, switching between different kinds of user representations to suit the current phase of the group collaboration. Lastly, they suggest that embodied avatars can be strategically placed beyond wall boundaries to support the user's workspace awareness within groups too large for the user's local environment. However, such a solution makes it more challenging to support the user's ability to spontaneously change their position within the group formation through literally embodied interaction.

Irlitti *et al.* (2023) explore cross-modality collaboration in hybrid group formations. Their work suggests that **multi-modal integration is crucial**; when users have different representations, with some only partially embodied, it creates communication asymmetries that lead to incongruencies in the experiences across the group. For instance, while unencumbered users might be able to hear remote users without seeing them, they do not have sufficient means to be able to respect their personal space (Irlitti *et al.*, 2023). This makes the support for dynamic group formation and conversational turn-taking more challenging.

While this grand challenge complements C1 and C2 (i.e., spatial heterogeneity and user representation), it focuses on the *interplay* between users and spaces and how distributed interpersonal space dynamics can support group formations.

Given the small body of research in this area, there are considerable challenges that remain to be addressed. We propose the following questions as a start:

- Technique Designing interaction techniques to support dynamic avatar transitions:** As large-group avatar interactions may occur across physically constrained spaces, alternatives to full-body avatars must be considered. One potential avenue for accommodating larger group sizes is to use dynamic user representations, such as switching between embodied avatars (e.g., Meta Avatars), disembodied representations (e.g., Apple’s flat FaceTime avatars), or partial embodiments (e.g., Apple’s spatial personas). These different avatar embodiments offer different trade-offs for managing larger group interactions and fitting people in confined spaces, but several questions arise. *Upon what criteria might avatar representations be adapted? (e.g., based on roles, physical proximity, participation level, etc.) How might users interact with each other when represented differently? How will the switching impact the overall collaborative experience?*
- Empirical and Technique Investigating spatial zoning techniques:** While many techniques for mapping features of space across locations have been studied (see C1), the ideas have not been tested at scale. *To what extent can techniques proposed in prior work support users in effectively navigating group dynamics at scale?* Answering these questions through empirical studies would involve iterating on dyadic technique designs to scale to distributed group spaces, e.g., using smaller local zones retrofitted to functional areas of a larger space.
- Design space Leveraging multimodality for distorted group spaces:** Most prior work focuses on the visual modality. However, to support spatial congruency at scale (for larger numbers of people and spaces), solutions must work across multiple modalities. For instance, as group size increases, we increasingly rely on spatial audio for turn-taking (Hyrkas *et al.*, 2023). As such, we must consider the problem across modalities. An initial step

here may be to develop a design space of multimodal cues: *How can multimodality be leveraged to help the users navigate group dynamics? How can avatar and space distortions be supported beyond the visual modality?*

7

C4: Live embodied conversations through asynchronous exchanges

With increasing globalisation, distributed teams can now involve members separated by space, time, or a combination of both. Asynchronous exchanges largely focus on work undertaken at different times sharing the same physical space (such as shift-work, process logging, etc). However, if we are to reduce the need for travel, solutions must also address how collaboration can be sustained even across locations and with limited overlap of working hours (e.g., due to time zone differences). As we have covered earlier, modern work is adopting new tools for asynchronous collaboration, but with similar issues to video conferencing. In this context, MR offers the potential to bring back the richness of users' embodied representations to these asynchronous interactions, enabling a new form of co-presence **across time** and space. In addition, MR supports spatially grounded asynchronous collaboration, whereby participants can incorporate elements of their local context into their interactions. This goal can be expressed as follows:

Grand Challenge C4: *Each collaborator should be able to engage in **live embodied conversations** around task objects **across time**, allowing them to both review and contribute to past conversations for future decision-making.*

7.1 Research challenge

To achieve this outcome, we must extend embodied interactions to **enable immersive synchronous team conversations and shared spatial memories with collaborators who are currently absent**. An example scenario of such interactions is illustrated in Figure 7.1.

In this scenario, both Alice and David can benefit from Bob's embodied communication cues during his breakout group *in the past*. This experience allows for the understanding of not only his verbal utterances but also his associated embodied responses. These conversational interactions are created using traditional synchronous face-to-face exchanges. The challenge becomes how to differentiate between explicit conversations with future or past peers and those who are in the current moment. Further, when a visual transition into the environment of a past meeting occurs, the experience must feel anchored to the physical reality of those in the new conversation. There are many challenges involved in fully leveraging this potential.

7.2 State-of-the-art research and open questions

Though many papers have explored the possibility of embodied asynchronous collaboration in VR (e.g., Liliya *et al.* (2020), Hubenschmid *et al.* (2022), Wang *et al.* (2024), and Bai *et al.* (2025)), few have dealt with the issues created by the physical space in MR (e.g., Fender and Holz (2022), Müller *et al.* (2021), Irlitti *et al.* (2016), and Cho *et al.* (2023)). The idea of leveraging the socio-spatial context for re-watching meetings was explored by Wang *et al.* (2024) and Bai *et al.* (2025), recognising the challenge of integrating absent users into a VR meeting recording. However, because of the flexibility afforded by VR, Wang

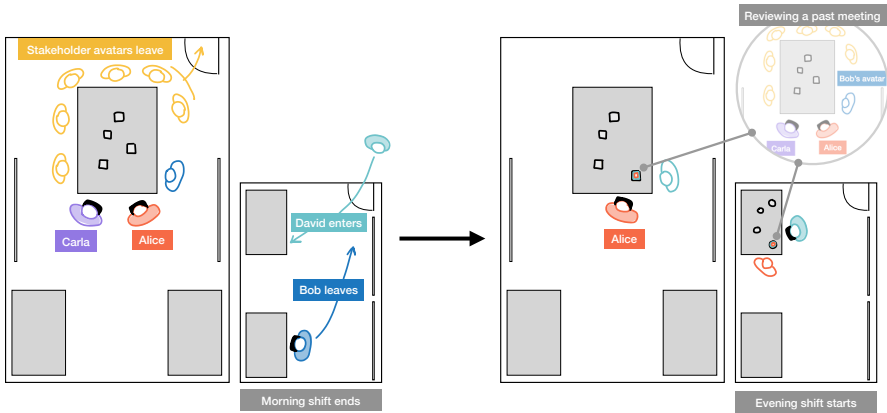


Figure 7.1: *Live embodied conversations across time:* Alice and Carla work regular hours, whereas Bob and David are shift workers; Bob has the morning shift, and David has the evening shift. After the stakeholder meeting, as Bob finishes his shift, he provides a quick handover to David and then leaves for the day. A short time later, as Alice and David review the feedback from the plenary session, they notice that Bob completed some amendments that do not align with their expectations. Bob will not be available until the following day to provide guidance. Therefore, using the physical building model as a point of temporal and spatial congruency, Alice and David load Bob and his breakout group’s session into their distributed meeting. For the next twenty minutes, Alice and David interrogate the communication and interactions between Bob and his stakeholder peers to understand the reasoning behind the changes, leaving their own embodied responses in keyframes for Bob to review the next day.

et al. (2024) could, for instance, adapt the behaviour of the meeting participants to make those watching it feel more included. Such adaptations included adjusting the f-formations of the group to incorporate the new participant and adjusting the original participants’ gaze behaviour to also acknowledge the new participant. When translating these experiences from a VR context to MR, we must not only adapt the recording to the new social relationships created by the new participant but also

to the current physical environment around them (see C1).

Next, when exploring the **composition** of an asynchronous experience, the interactions can be viewed as either an assembly of *turn-taking*—the simpler case or dyadic collaboration—or *bichronous collaboration*—the more challenging case of distributed team collaboration with a hybrid of synchronous and asynchronous interactions.

In **turn-taking** interactions, all users participate asynchronously and contribute one at a time. For example, Fender and Holz (2022) presents a compelling vision of asynchronous workflows in MR where users can record spatially anchored point-cloud messages that others can later discover and view. An interesting challenge identified by these authors is presenting a causally coherent sequence of events. Maintaining a model of these sequences can also facilitate the temporal navigation of these recordings as the objects themselves can act as triggers for recordings that involve them, as demonstrated in VR by Liliya *et al.* (2020). In the more challenging case of **bichronous collaboration**, distributed participants interact synchronously while others interact asynchronously (Figure 7.1). By the time the absent participant engages, a whole conversation has already happened—a concept that has been explored in VR (Wang *et al.*, 2024; Bai *et al.*, 2025; Giovannelli *et al.*, 2025) and in AR Cho *et al.*, 2025b; Cho *et al.*, 2025a. However, state-of-the-art research has only scratched the surface of the problem, focusing on collaboration across different *times*, while assuming that collaboration happens in the same homogeneous *space* (either virtual or physical).

In such scenarios, the notions of interaction, visualisation, and perception must be revisited. While the playback of time can be controlled, time itself does not stop. How does one interact and contribute towards a past recording while maintaining the underlying structure of previous conversations? Enabling users to consume conversational exchanges efficiently without necessarily rewatching the entire original recording is an issue that compounds as more and more asynchronous iterations of the same meeting occur over time, creating challenges of the kind usually solved by version control systems like Git. Girgensohn *et al.*'s *Hypermeeting* system shows a potential solution to this problem in the context of videoconferencing. This system creates hyperlinks between subsequent

iterations of asynchronous meetings (Girgensohn *et al.*, 2015). The present grand challenge could lean on past examples of asynchronous collaboration concepts, such as creating threads or branches. However, complexity is in itself a prominent challenge. This would require ways of navigating through the recording (e.g., 3D timelines inspired by the field of immersive analytics (Hubenschmid *et al.*, 2022)).

As temporal interactions in immersive systems are an emerging area of research, there are a number of opportunities for future research to inform how MR can enable collaboration despite temporal disparities.

- Technique Anchoring past meetings in physical space:** MR presents novel opportunities for interaction with time, such as revisiting past decision-making processes (e.g., Irlitti *et al.* (2016), Cho *et al.* (2023), and Hubenschmid *et al.* (2022)), or augmenting situated spatial memory (e.g., Liliya *et al.* (2020)). However, past meetings may be revisited in and across physical environments that are different from where they were initially recorded. This poses challenges for maintaining consistent semantics of anchors within a team, bringing a new level of complexity to designing interaction techniques with time. *How can time be incorporated as a design material to augment spatial memory across heterogeneous spaces?*
- Theory and System Navigating increasing spatio-temporal complexity:** While temporal communication is simple between turn-taking dyads, the case of distributed teams in bichronous collaboration is more complex and under-explored. Our distributed workshop scenario (chapter 3) demonstrates how the notion of time can be blurred. Assuming that experiencing past meetings can be as immersive as the present, it can be said that Alice and Bob exist in two timelines; while they explore a past meeting, their interpersonal communication also exists in the present. *Which visualisations and interactive affordances should be provided to allow distributed teams to seamlessly transition between the past and the present?*

8

C5: Tangible interactions with objects across locations

MR has the potential to enable the ad hoc retrieval and incorporation of **tangible objects** into conversations – with a consistent view of the real world, users can continue to see and select from the local physical objects around them. However, the physical differences in objects and their locations can make continued, synchronous physical participation difficult for remote users.

Grand Challenge C5: *Every collaborator can communicate through spontaneous, shared interactions with **tangible** objects, with seamless replication and synchronisation of object state **across locations**.*

8.1 Research challenge

While much prior work has focused on asymmetric remote collaboration (Fidalgo *et al.*, 2023b; Huang *et al.*, 2022; Kim *et al.*, 2020), this challenge is about **enabling bidirectional interactions with tangible objects through digitally mediated communication**. The value proposition

of this capability is that both sides (in the above scenario) can leverage aspects of embodied cognition and interaction to support the ongoing discussions. To achieve this outcome, we must address (i) **selection**: how objects on each side are initiated into the collaboration, (ii) **physicality**: the physical properties of these objects, and (iii) **movement**: changes to their spatial location. Consider the scenario illustrated in Figure 8.1:

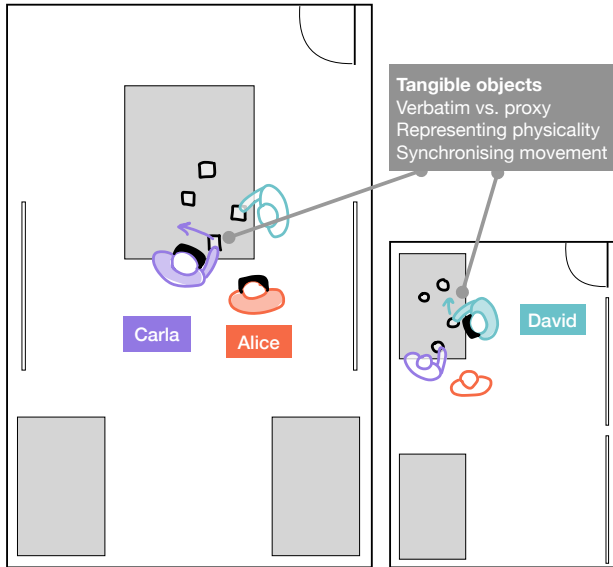


Figure 8.1: *Tangibility challenge*: After the stakeholder workshop, Alice, Carla, and David stick around to update their urban plan. To brainstorm and visualise the positioning of buildings, Alice and Carla place the physical scale models of buildings on their local table. As David does not have local access to the physical models, he reaches forward and grabs the salt, pepper, and two glasses to represent the buildings. The scale model and proxy objects are moved as spatial referents to support their urban planning. Any of the architects can reach forward and make physical manipulations to the setup, to incorporate their ideas.

In this scenario, Alice and Carla leverage scale models of the buildings to share ideas and find solutions. Here, each object is represented literally, and **verbatim**. Alice and Carla, therefore, can benefit not only

from the objects' spatial relations but also from their passive haptics. David, conversely, must leverage objects that are available in his space. Fortunately, he has available objects that he can *select* from. These objects are ad hoc, simple, and available **proxy** objects. For him, their *physical* form has no value, but their spatiotemporal configuration is key. His interactions with the objects are necessarily less rich than Alice's and Carla's. While this interaction may seem asymmetric, the crucial feature of symmetric tangible interactions is that all collaborators can participate in thinking through doing – i.e., the physical objects can serve as the tools and materials of the collaboration.

8.2 State-of-the-art research and open questions

The challenges around selecting appropriate proxy objects, mismatches in their physical details, and the inability to physically move tangible objects across distributed spaces are fundamental gaps that make physical tasks highly challenging in distributed settings.

Collaborative MR systems must support **selecting** and **adapting the physicality** of proxy objects. A line of research explores ways to make object selection opportunistic and spontaneous based on object and gesture recognition (e.g., Henderson and Feiner (2008), Du *et al.* (2022), Hettiarachchi and Wigdor (2016), and Greenslade *et al.* (2023)). To then leverage the physicality of these objects, research in VR has been examining solutions to differences in objects' physical properties through visuo-proprioceptive illusions (e.g., (Bergström *et al.*, 2019; Simeone *et al.*, 2015; Azmandian *et al.*, 2016)), and these may prove valuable for MR, too. The idea of leveraging both verbatim and proxy objects already exists in VR. Verbatim objects, for example, could be precisely tracked and represented by an exact physical model, allowing them to be represented one-for-one with their physical counterpart. Proxy objects can work similarly simply by switching their visual representation for a different model. In both scenarios, the VR user can reach out and incorporate these objects into their peripersonal space and interactions. This is simpler in MR, as the user can directly and constantly see the space and objects around them. If realistic rendering and compositing can be achieved, then proxy overlays may work seamlessly, too. Steps

towards this have been shown through diminished reality (Cheng *et al.*, 2022), for example. Overall, the challenge arises in whether a physical proxy object can be made to feel like a different virtual object.

The haptic community has been working on this problem, primarily in VR, through two approaches: active haptics and passive haptics. Active haptics explores mechanical, shape-changing, specialist devices that seek to accurately recreate specific haptic details (e.g., size (Gonzalez *et al.*, 2021), shape (Benko *et al.*, 2016), and slippage (Kim *et al.*, 2022)). Passive haptics, conversely, pursue illusory, visual-proprioceptive manipulations to leverage random physical proxy objects as haptics for their virtual counterparts. In this area, research has shown how physical objects can represent objects of different sizes (Bergström *et al.*, 2019), shapes, and curvatures (Tinguy *et al.*, 2019). To date, our haptic solutions remain relatively specialist and low-resolution. More general approaches and grounding theory are needed to drive this area forward.

Finally, a major challenge arises as Alice and Carla **move the tangible objects** around. David’s proxy objects become physically dislocated from their virtual positions, decoupling David from the socially situated practice. Illusory solutions from passive haptics research, for example, allow us to account for small discrepancies between a virtual object’s location and the location of its physical counterpart. We can use these illusions to lightly redirect the user’s hand from the virtual object’s location to the physical object’s location as they reach forward. The scope of this redirection, however, is limited before the user becomes aware of the manipulation and their experience degrades dramatically. Research on these illusions reveals a ‘haptic coverage’ of only ~ 15 degrees to the left or right (Clarence *et al.*, 2022), or $\sim 0.87 - 1.31$ x the distance to the virtual object (Esmaeili *et al.*, 2020).

Moving beyond small-scale redirections, supporting larger remote movements of physical objects requires more substantial, typically robotic solutions. Prior research has demonstrated tangible interactions at different scales. This includes adopting “robot-coasters”, leveraging the ideas presented in Zooids (Le Goc *et al.*, 2016) to carry objects around a surface. In this approach, tangible interactions are synchronised to provide congruency in the spatial configuration across distributed locations (Auda *et al.*, 2021; Richter *et al.*, 2007). To scale up to larger

objects, prior work has proposed general-purpose shape-changing structures built into furniture that can reconstruct the tangible shape of objects, such as inFORM (Follmer *et al.*, 2013). With subtle mechanical tweaks, such a table may be able to move more objects around a space and represent objects themselves (Ishii *et al.*, 2015). At room scale, we could imagine the synchronisation of dynamic physical features of space (Suzuki *et al.*, 2020; Grønbæk *et al.*, 2017) and even drone-based haptic devices for mid-air interactions (Abtahi *et al.*, 2019). However, these all begin to require substantial infrastructuring or physical manipulations that become more complicated to disappear into the background.

To achieve bidirectional, distributed interactions with tangibles, we outline critical next steps:

- **Technique Selecting physical proxies:** Most work on remote tangible collaboration assumes that the tangible interaction is set up a priori. However, while in face-to-face collaboration, we can spontaneously grab and incorporate nearby physical objects, distributed settings require that this interaction is *initiated* (e.g., Hu *et al.* (2023)). Object interactions may thus not be seamless for distributed users, leading to open questions for interaction technique design. *Which approach serves best to mediate spontaneous object interactions? Should a selection interaction be integrated at every stage? Should they begin with visual-only, non-physical representations, or does this serve to exclude them from the socially situated practice? Should the system prompt them immediately to find a physical proxy object?*
- **Technique and Empirical Retargeting illusions in multi-user settings:** To date, much of the illusory research conducted by the haptics community has considered only single-user scenarios. This is appropriate for David, using physical objects on his own. For Alice and Carla, however, who are co-located, using these illusory techniques would necessitate the introduction of visual discrepancies between Alice and Carla. The impact of this on their trust, experience, and social presence remains an open question.

How can retargeting illusions be extended as multi-user interaction techniques? And how do such techniques affect the individual and collaborative experiences of interacting with real objects?

C6: Fluid interactions in complex ecologies of devices and applications

While MR headsets may converge in their capabilities (Ens *et al.*, 2019), their success likely depends on how well they integrate with people’s existing artefact ecologies (Bødker and Klokmoose, 2012; Buxton, 2010), such as devices, digital applications, and physical artefacts they use for work. Therefore, it is unhelpful to view MR as a replacement for all other devices. Rather, it may complement people’s existing tools—and as we explore in this manuscript, one of the headset’s primary functions is likely to be to access embodied meeting experiences for distributed teams (Ens *et al.*, 2019). To make the MR headset truly complementary, it must first work fluidly together with the existing ecology of devices, such that it supports easy transitions between tools and tasks. We use the term *fluid interaction* to describe the ability to use technology for accomplishing a task without allocating excessive attentional resources to the mechanics of the technology. According to Elmqvist et al.’s 2011 operational definition, fluid interaction promotes flow, direct and embodied interactions, consistency between the actual and user-perceived states, and use of tools and objects. Such interactive experiences are especially difficult to achieve when considering MR users working across multiple devices and applications simultaneously.

As such, this part of the agenda focuses on the collaborative context of **tasks** and the tools we use to solve them:

Grand Challenge C6: *The collaborators can **fluidly** transition between collaborative tasks, navigating the **complex** ecology of task items to spontaneously share contents both within MR and between MR and other modalities.*

9.1 Research challenge

Fluid interaction is often presumed in the physical realm, where one can directly switch between a stack of sticky notes on the desk to scribbles on the whiteboard. In the digital realm, we have come a long way, e.g., enabling fluid creation and editing of digital documents, synchronised across devices with different ecosystems, and easily shared with collaborators. An extensive body of research has investigated the HCI challenges related to cross-device interaction (Brudy *et al.*, 2019).

However, the more challenging situations are when tasks must **transition between the digital and physical realm**. As an example of today, people often take pictures of their whiteboards to have them on their computing devices. For this purpose, MR has a unique potential **to serve as a “bridge” device**. Consider David’s trajectory of work as he enters his shift, illustrated in Figure 9.1.

When comparing this envisioned scenario to the current consumer technology landscape, it becomes clear that significant advancements are still needed. Available MR headsets do not come with mature infrastructure to support fluid interactions between available applications, different physical set-ups, and various collaborative tasks. For instance, switching applications within MR for different tasks hinders fluidity due to lacking support for shared spatial maps to maintain digital state across applications. Another example is the switching cost between virtual overlays and reality, such as issues with attentional tunneling (Syiem *et al.*, 2021) and task-refocusing (Lystbæk *et al.*, 2024) preventing effective task-switching in MR environments. Additionally,

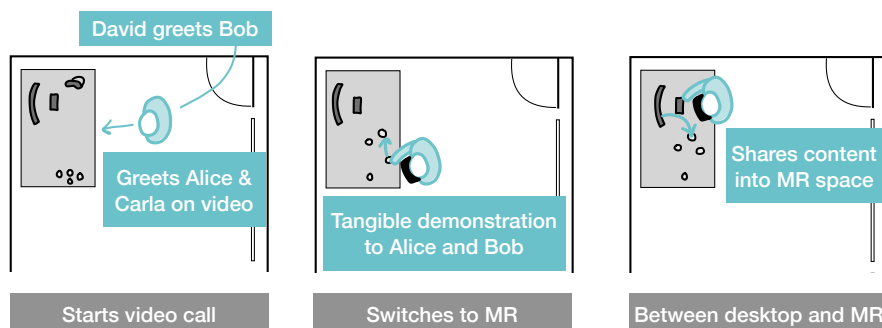


Figure 9.1: *Fluidity challenge*: This illustration follows David’s individual trajectory of work as he engages with his colleagues. First, David enters the doorway to the office, initiating a greeting with Bob, who is co-located. Bob’s shift has ended, and David takes over by joining the workshop debriefing with Alice and Carla through a video call. Second, as they switch focus to the urban plan, he puts on his MR headset to see it spatially in front of him. He brings in the tangible proxies to demonstrate a spatial layout suggestion to the others. Alice and Carla struggle to follow his idea, and he turns to the whiteboard to illustrate it in the abstract. Third and finally, before they wrap up, David shares a new material rendering that he has produced on his desktop computer. With a pick-and-drop, he moves the texture from his desktop display onto the selected proxy object in front of him to fluidly share content across devices.

non-standardised interaction paradigms and capabilities further hinder fluid interactions between collaborators using different MR headsets. Part of what made the Vision Pro useful on day one is the integration and continuity within Apple’s ecosystem ¹. However, this is then locked to Apple’s proprietary solutions. There is currently no universal infrastructure with open cross-platform standards, and the lack of integration across people’s ecologies limits the potential for consumer adoption of MR for work. It is thus imperative to design MR solutions that account for the changing tools, tasks, and devices in contemporary collaborative

¹The Street review of Apple Vision Pro <https://www.thestreet.com/technology/apple-vision-pro-review>

work, if we are to support fluidity.

9.2 State-of-the-art research and open questions

Only recently has research started to engage with how MR headsets might work together with existing multi-device ecologies. A research community is currently emerging under labels such as cross-reality interaction (Schröder *et al.*, 2023; Auda *et al.*, 2023), and complementary (hybrid) user interfaces for MR (Hubenschmid *et al.*, 2023). However, this field is in its infancy, and while a few works seek to understand how collaborating users might transition between headsets and other devices (e.g., Schröder *et al.* (2023)), there is little investigation of how MR technology can support fluid interactions to initiate collaboration contexts *ad hoc*, and switch between tasks and tools that span across the physical and digital realms.

As such, the outlook for this challenge requires fundamental research into questions around how MR headsets might work, not in isolation but in conjunction with existing ecologies of practices and tools for distributed collaboration.

- **Technique Ad hoc cross-reality interaction techniques:** Most prior work on MR cross-device interaction techniques assumes a priori pairing between devices, yet techniques that allow transitioning and sharing to be initiated *ad hoc* are under-explored. *How can we design fluid interaction techniques that support starting and joining meeting contexts with low friction? How can a video call turn into an MR meeting? How can users share content across digital and physical realms, such as porting physical whiteboard contents or sharing files from their desktop computer into an MR collaboration context?*
- **Toolkit Cross-reality open-source toolkits:** This emerging research community would benefit from toolkits to accelerate the development of techniques for cross-reality prototyping. While proprietary solutions are being developed for this², there is currently

²Apple continuity features <https://support.apple.com/en-us/108046>

a lack of mature open standards. WebXR³ provides a promising low-level standardised infrastructure. However, open-source HCI toolkits can provide protocols at a higher level, closer to the end-user. As such, technical HCI work should look into this question: *How can we support rapid prototyping, standardisation, and reuse of interaction techniques for cross-device MR systems?*

- **Empirical and Methodological Living Labs to study the longitudinal role of MR:** There is a lack of methodological innovation in how we study the use of MR, and recent technology maturity paves the way for running participatory studies with stakeholders to explore questions of early adoption in teamwork practices: *When is there a sufficient value proposition for switching to MR headsets for everyday work? How frequently will MR meetings be needed?* However, the current methodological challenge is: *How do we design studies to get a glimpse of the answer to these empirical questions?* We currently have a “chicken and egg problem”; even if there were collaborative MR applications and cross-reality interaction support, it would rely on the ability to convince teams to adopt MR headsets for their real distributed meetings. However, the first step here might be to adopt a Living Lab methodology (Alavi *et al.*, 2020). Famously, researchers pioneering Media Spaces took a similar approach by adopting video-conferencing for their distributed collaborations across research labs, decades ahead of mass consumer adoption (Bly *et al.*, 1993; Harrison, 2009).

³The WebXR standard <https://www.w3.org/TR/webxr/>

10

Conclusions and Future Outlook

We have offered a comprehensive account of some of the grand challenges ahead for research and innovation to truly enable distributed team collaboration in Mixed Reality. To provide an outlook for future work in this area, we chart future directions posed by the challenges and discuss how to approach them as a collection of concerns.

10.1 From vision to reality: an inventory of research questions

We have taken stock of the state of distributed MR collaboration research from the perspective of current events, including recent MR technology progress and changes in work practices in knowledge work as well as open questions for future work associated with each grand challenge. We have demonstrated their relevance by highlighting their timeliness in modern distributed work, and we have demonstrated that they are under-explored by identifying gaps in state-of-the-art literature.

The purpose of framing challenges through these statements of tension is that they explicate the compromises that designers must navigate. Thus, we see two primary ways that this manuscript can be used to inspire action for future research. First, it provides an inventory of identified research questions, highlighting specific future contributions

to fill research gaps within the scope of the grand challenges to make distributed MR like face-to-face collaboration. Second, the manuscript embodies the methodological approach of identifying tensions when a vision is faced with reality, in this case, revealed at the intersection between face-to-face principles and modern work requirements.

We hope that this approach will inspire future endeavours to continually challenge the community's MR visions by identifying new tensions or updating them as modern work requirements continue to evolve.

10.2 How the grand challenges address well-known concerns for modern work in a timely manner

Timeliness is a central concern in this manuscript. Ens *et al.* (2019) argue that we need to develop conceptual frameworks to further identify uncharted areas for research on collaborative MR. While the grand challenges C1—C6 (Figure 1.2) are situated in well-charted research areas, their sharp focus on tensions with today's world of work shines new light on these concerns.

Let us review how each challenge statement offers a fresh perspective on its area of research. The challenges of heterogeneity across spaces (C1) and devices (C6) capture emerging subfields (spatial heterogeneity (Wong *et al.*, 2025) and cross-reality interfaces (Auda *et al.*, 2023)) which are still in the midst of establishing common vocabulary and design principles. Challenges in diversity of user representations (C2) received recent attention in the context of hybrid collaborative configurations, which remain a cutting-edge issue (e.g., Bjørn *et al.* (2024) and Irlitti *et al.* (2023)). The problem of scaling distributed MR to group dynamics across space (C3) and time (C4) are mostly explored in VR and under-explored in MR. Finally, while research on symmetric multi-user tangible interactions (C5) started decades ago, the recent advances in novel techniques for proxies and haptic retargeting allow us to revisit it with new approaches. The premise of this agenda is that, when a modern distributed team needs principles of face-to-face interaction, the incentive today is to travel large distances for in-person workshops and meet-ups—and this is not a sustainable practice. Despite the timely nature of this argument, it is likely to persist for as long as the alternative

solutions continue to miss the essential benefits of face-to-face interaction. Yet, we encourage future work to continually update and extend the agenda to remain relevant for contemporary work practices.

10.3 Interconnectivity of the challenges in modern work settings

Collaboration in everyday work environments is a dynamic process which can involve switching between interactions addressed in the challenges. As a starting point, our manuscript distils six grand challenges as individual elements, so as to clearly articulate a research agenda. However, the interactions developed to address these grand challenges will ultimately be applied by users in a way that interweaves aspects of all six dimensions.

Future work should consider the intersection of multiple challenges and how the solutions for these might influence each other and produce new problem spaces or phenomena. For example, there may be scenarios where large groups of participants need to be situated across spaces with varying degrees of capacity. As a consequence, user representations (C2) need to be adapted so that these large groups of people (C3) can be rendered coherently across the various heterogeneous spaces (C1). Combining these challenges begins to surface edge cases and new exciting opportunities for MR solutions that go beyond the immediate agenda proposed in this manuscript.

10.4 Beyond the six grand challenges

Our manuscript is by no means a checklist that needs to be addressed before researchers can explore solutions that go “beyond being there”. Rather, it identifies the fundamental areas of friction that circumvent a future where the full potential of MR is realised. Interactions that go beyond being there are still a valuable endeavour. However, we posit that as long as the available MR meeting interfaces are far from on par with the cognitive ease of face-to-face interaction or the fluidity of setting up video-conferencing calls, MR will remain a gimmick rather than a tool for effective collaboration. Similar to how Canfield-Smith’s (Smith *et al.*, 1982) desktop metaphor helped the general population

understand abstract graphical user interface concepts (like storing files on a computer), MR interfaces should help their users intuitively engage with its fundamental principles of interaction. Yet unlike the desktop metaphor, many of the interactions in MR are designed to be situated in the real world. Here, metaphors melt away (Wong *et al.*, 2024b) and we are left with a set of fundamental, familiar interactions that should simply look and feel like face-to-face collaboration—to the degree that the distributed team’s work arrangements allow for.

As we advance on these grand challenges, it is crucial to continually reflect on the implications of the solutions that are proposed. This includes ethical considerations such as how it might change incentives of work (e.g., by making it even easier to not leave the bedroom for work). There are also critical privacy concerns with MR sensing technologies used from the privacy of the home. Finally, there are issues around agency and trust in the systems, especially if we offload much of the embodiment rendering and spatial re-mapping to an intelligent system. To what extent does it matter that people are aware of the system altering their distributed collaboration spaces and the actors therein? What kinds of additional benefits could altering of shared spaces provide that go beyond traditional face-to-face principles? E.g., how could alterations be designed to make collaborative MR spaces more accessible and inclusive?

10.5 Final remarks

While the vision of MR for distributed team collaboration has been around for decades, it has lacked a clear articulation of how it can be realised in the real-world context of distributed work. In this manuscript, we have brought together theories and recent studies of CSCW to articulate grand challenges for how to make it a reality. Recentering the agenda on grand challenges derived from tensions—between key face-to-face principles and requirements of modern work—helps clarify not only design trade-offs but also the next open questions for future research.

The manuscript offers an agenda of grand challenges enabling researchers, designers, and other decision-makers to ground the discussion

in tensions between *human face-to-face skills* and *contemporary work demands*. Acknowledging and exploring the interaction design trade-offs through this lens is imperative to deciding how MR solutions can best reconcile the tensions between the two. With this research agenda, we thus make a timely call to action toward accelerating research on MR, which we believe to provide a more sustainable paradigm for distributed team collaboration.

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